

investor withdrawals.¹³ What we need is an indicator of maximum adverse consequences so we can avoid strategies that have too much left tail risk.

One such indicator is conditional value-at-risk (CVaR), also known as expected shortfall. CVaR uses the actual distribution of returns to determine the expected loss of a portfolio when there is a loss. CVaR is difficult to calculate, and the results are not intuitively appealing. I find it difficult to relate to the CVaR values and prefer instead to use a visual indicator called a box plot. This shows on one comparative chart median returns, interquartile ranges of returns, and expected extreme values.

Another simple indicator of tail risk that is intuitive, easy to understand, and relatively easy to calculate is maximum drawdown.¹⁴ Drawdown is the percentage that price moves down from a new high. Since we use monthly returns, maximum drawdown to us means the maximum cumulative peak-to-valley retracement on a month-end basis.¹⁵

As with most things, there are some potential drawbacks to using maximum drawdown. First, maximum drawdown is dependent on the length of one's performance record. All else being equal, maximum drawdown increases with track-record length. Therefore, it is most useful when maximum drawdown is used to evaluate strategies having the same amount of performance history and plenty of historical data. Second, maximum drawdown represents only a single occurrence. The number of drawdowns that occur and drawdowns other than the very worst one may also be important to us. To get a better sense of the depth, quantity, and duration of drawdowns, I look at drawdowns in different ways, at different times, and under different conditions.

AN INTEGRATED APPROACH

We are now ready to evaluate our strategies numerically and visually using comparative returns, standard deviations, profit consistency, alphas, Sharpe ratios, box plots, and maximum drawdowns under different scenarios.

We have looked at the history and evolution of momentum, the rationales that support it, and what we should or should not include as investment assets. Now that we have the requisite background information